Data augmentation with Generative AI for DoW attack detection in serverless architectures

Mora, H.; Ortega Candel J.M., Mora Gimeno F.J., Fernández Zorrilla, M.
Department of Computer Technology and Computation
University of Alicante

Abstract.-

Serverless computing is one of the latest paradigms in cloud computing. It offers a framework for the development of event-driven applications whose functions are executed in a scalable environment provided by the corresponding cloud platform. In this way, resources are obtained on demand, paying only for the time the function is running. This new model has new vulnerabilities and, therefore, new types of cybersecurity attacks. However, there are still not enough transaction datasets for serverless systems with a sufficient amount of data to develop advanced detection methods for this type of threat. Therefore, we present this dataset that has been built with generative AI to advance the development of models that can effectively deal with these threats.

Keywords.-

Serverless computing, Generative AI, cloud architectures, cybersecurity, DoW

1. Introduction

Serverless computing is a deployment architecture model that aims to provide functionality in the cloud in such a way that the execution of some functions of a distributed system takes place on a server of the corresponding cloud provider. Thus, the user of this cloud service pays based on the execution time of these functions on the cloud server. Thus, the developer who wants to deploy his application in serverless mode, agrees a budget that will be available as a balance and that will be consumed as calls are made to the functions deployed.

The "Cloud Native Computing Foundation" (CNCF) defines serverless computing as the concept of building and running applications that do not require server management. It describes it as a fine-grained deployment model where applications, included as a set of functions, are uploaded to a platform where they are executed, scaled and billed based on the exact demand at that moment [1, 2]. This is why the name serverless can be misleading, since this model does not work without a server, but from the application developer's point of view, neither the acquisition nor the configuration of a server is necessary for the implementation of a networked application [3]. In fact, the first time the term serverless was coined was to refer to some peer-to-peer networks, since in these the concept of 'serverless' is more in line with the reality of a system where a set of nodes pretends to behave as a single computer, taking advantage of the resources of all of them.

A recent study by cloud monitoring platform Datadog found that the adoption of serverless environments by organisations running their software on Google Cloud and Azure had grown by 6 and 7% respectively over the 2022 results [4]. While those running on Amazon Web Services had grown by 3% and now account for more than 70% of the company's total.

Within the standardised service models in cloud technology, serverless is framed within SaaS (Software as a Service), the user does not control the cloud infrastructure, which includes network, servers, operating systems or storage. This is why the security analysis will focus on protecting communications and ensuring the integrity of the use of serverless functions [5]. Specially, when this model is used for IoT or edge applications [6, 7].

The Denial-of-Wallet (DoW) attack is an adaptation of the classic Denial-of-Service (DoS) attack. The principle of both attacks is the same: to exhaust the response capacity to a flood of requests to a service, making it impossible to use it properly. The difference is that in a DoS attack the aim is to render the service unusable, while in a DoW attack the aim is to exploit the limit of the computing capacity contracted with the cloud service, seeking to cause direct economic damage.

There are practically no datasets to study this treat. This is because of its relative novelty, although serverless systems have already been in place for some time, and on the other hand, because these datasets should be provided by cloud providers. Thus, there are only a few proposals with datasets to research on this field [8, 9].

2. DataSet Generation

In order to build sufficiently robust artificial intelligence models, it is necessary to have a solid foundation, which requires a data set or dataset that allows the model to train itself to correctly interpret the new data with which it will work.

A good dataset is the foundation on which a solid and reliable artificial intelligence model is built. By providing a representative variety of examples, a quality dataset allows the model to learn patterns and generalise correctly to new situations. In addition, the quality of the data directly influences the accuracy and efficiency of the resulting model. However, biased data can distort the training process and lead to erroneous conclusions. Bias can arise in different ways, such as a lack of diversity in the samples or the inclusion of human bias in the data collected. This can lead to unfair decisions or discrimination when the model is deployed in the real world, as it will reflect and amplify biases present in the data.

Artificial data generation is the process of creating data that mimics the characteristics of real data. It is typically used in cases where real data is limited, as there may be privacy concerns or data scarcity.

Data augmentation can help reduce overfitting by generating new data that can be used to train the model, making it more robust and generalizable [10, 11]. This increases the quality of the model's performance on real data, and therefore it will be possible to implement it in real applications working on an intrusion detection system.

Currently, the reference model for synthetic data generation and automatic dataset augmentation are the Generative Adversarial Network (GAN). The effectiveness of GANs lies in the competitive nature of their structure; two models are pitted against each other (hence ‘adversarial’) in an attempt to generate the most realistic synthetic data possible. To do this, the generator tries to develop new samples that are identical to the original ones, while the discriminator learns to discern the real data from the generated ones. This feedback results in the data that pass the discriminator's filter being data with sufficient value and fidelity to the original distribution. So they can be perfectly employed in the training and testing phases of the new improved models. Generator and discriminator are the two legs that support generative adversarial networks. This type of techniques based on deep learning, after all they are two neural networks networks facing each other, result in a powerful artificial intelligence capable of generating new data [12].

The quality of synthetic data generated by a GAN depends on the complexity of its generator and discriminator components, the quality of the training data and the number of iterations employed during the training process [13].

3. Dataset Description

The dataset generated contains calls to functions together with the event that has occurred, response times, duration of the functions, functions that are activated in the request and in the response, memory and CPU usage. Each function has the event that originated it. To give more relevance to the types of events, the dataset includes a column called **functionTrigger** that includes the following possible events:

functionTrigger { "http", "storage", "sql", "stream", "notification", "email","nosql"}

The following table shows the name of each column together with the data type:

Column	DataType
Id	int64
IP	object
bot	bool
FunctionId	int64
functionTrigger	object
timestamp	object
SubmitTime	int64
RTT	int64
InvocationDelay	float64
ResponseDelay	float64
FunctionDuration	float64

ActiveFunctionsAtRequest	int64
ActiveFunctionsAtResponse	int64
maxcpu	float64
avgcpu	float64
p95maxcpu	float64
vmcategory	object
vmcorecountbucket	int64
vmmemorybucket	float64

The generated dataset is characterised by the following data at the level of generated transactions:

Transacciones	**2000000**
Attack Transaccions	**100000**
No Attack Transaccions	**1900000**
Percentage Attack Transaccions	**5%**
Percentage No Attack Transaccions	**95%**

Transactions are distributed over a 24-hour period. In the following graph we can see how legitimate requests are made between 0.00 and 3 am, while those originating from a botnet are concentrated between 3 am and 5 pm.

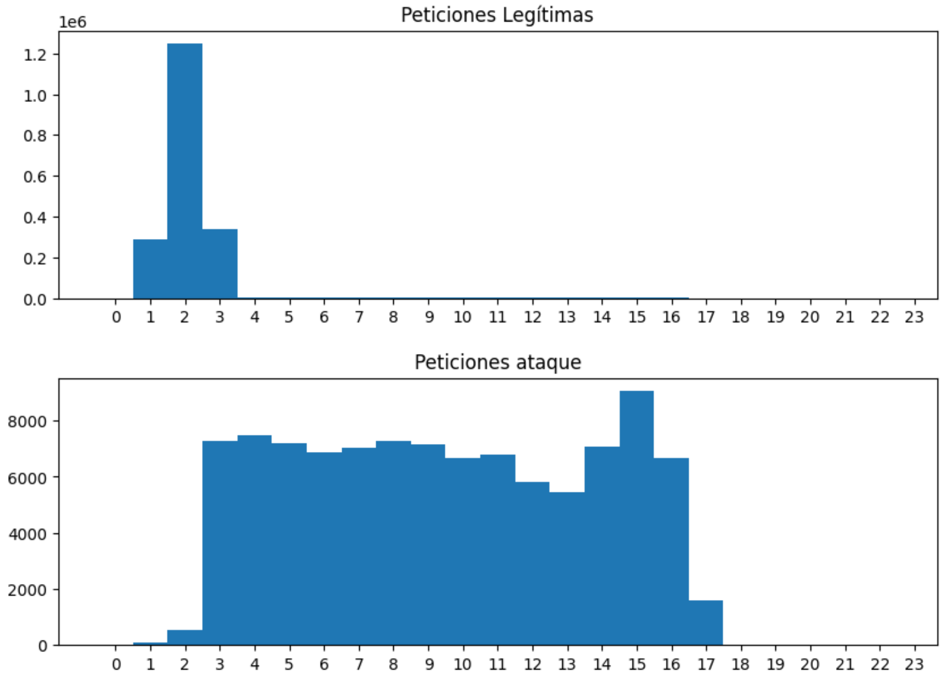

Fig. 1. Transaction distribution

The following graph represents a distribution of function identifiers (50 functions) on the x-axis along with the percentage of transactions generated by each function identifier on the y-axis.

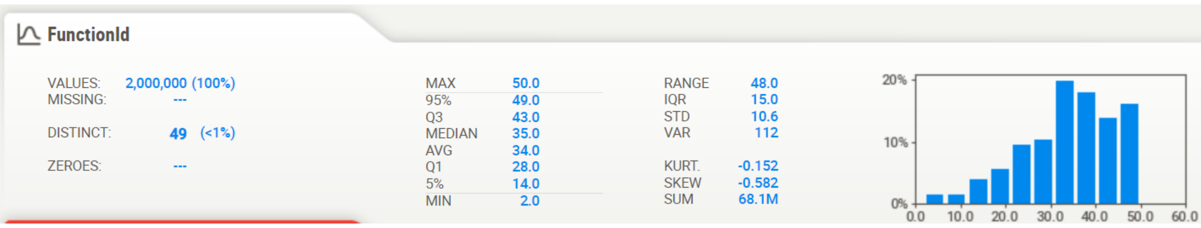

Fig. 2. Function distribution

The following graph represents a distribution of dispatch times on the x-axis along with the percentage of transactions generated on the y-axis.

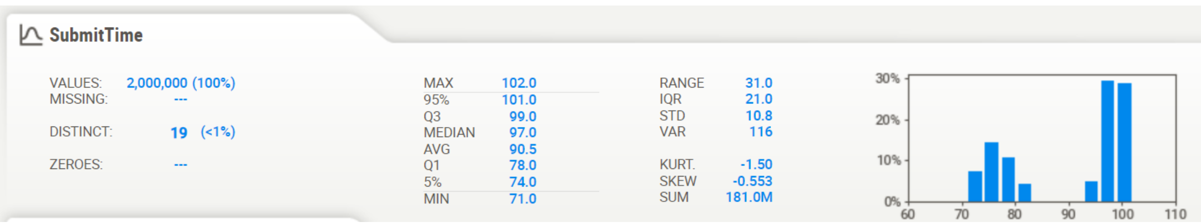

Fig. 3. Dispatch times distribution

The following table shows some statistics on attack and legitimate requests.

	No Attack requests	**Attack requests**

count	1.900000e+06	100000.000000
mean	1.031325e+03	1022.522770
std	1.045330e+01	15.067935
min	9.900000e+02	990.000000
25%	1.025000e+03	1012.000000
50%	1.032000e+03	1023.000000
75%	1.039000e+03	1036.000000
max	1.049000e+03	1049.000000

Correlation matrix

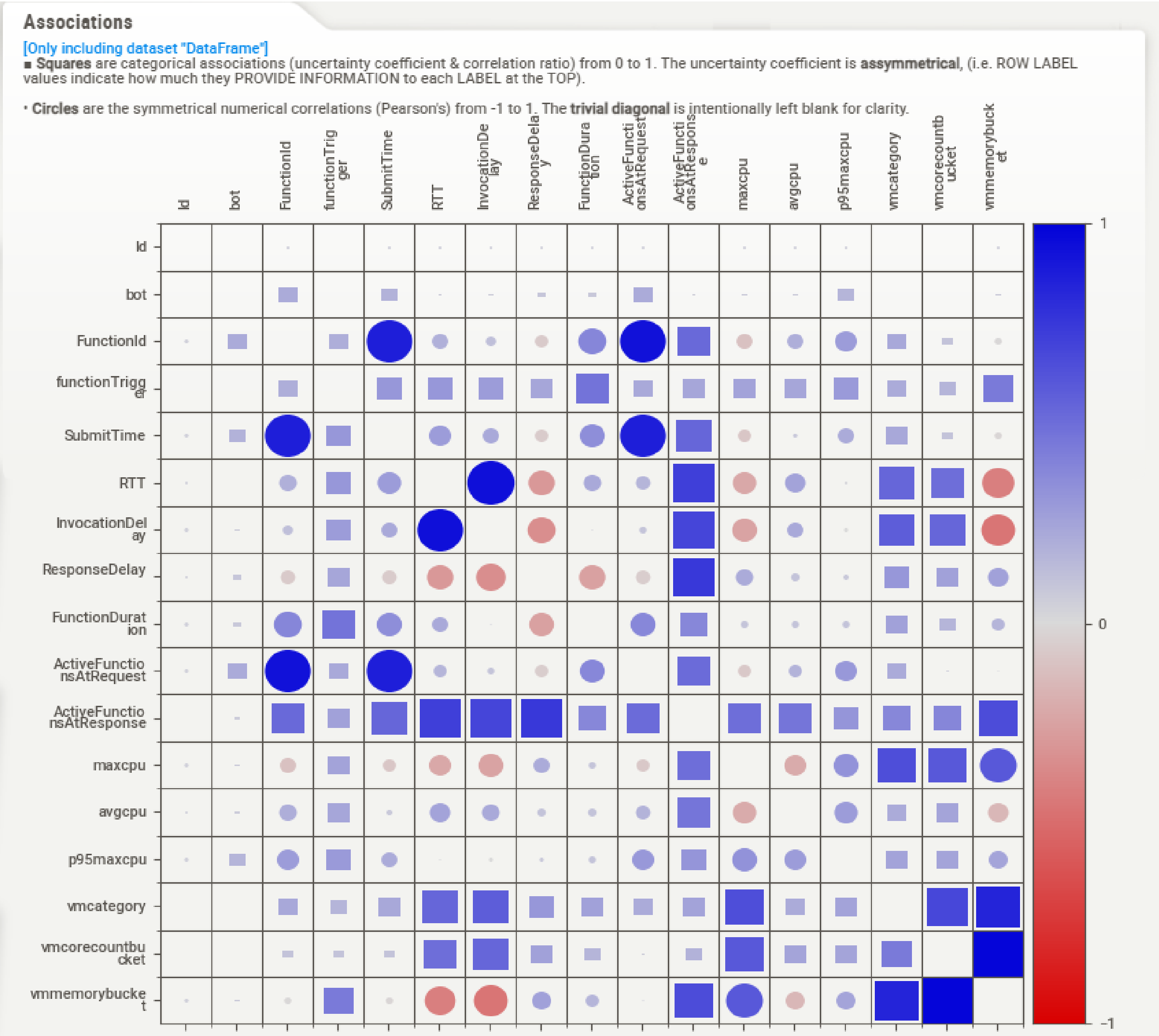

Fig. 4. Correlation matrix

Attack vs. non-attack transactions by event type

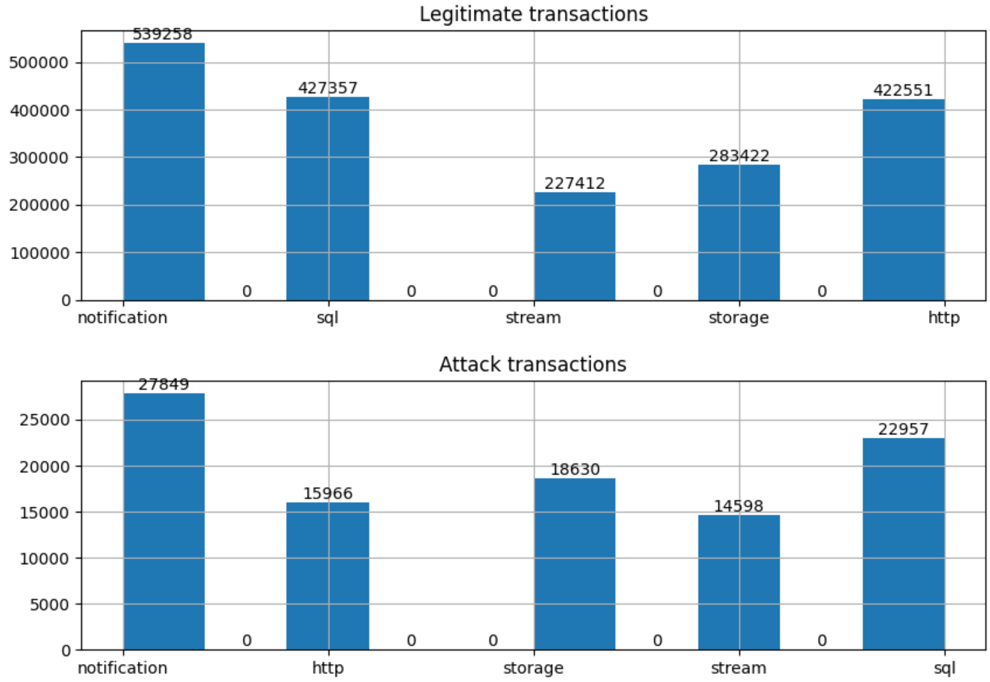

Fig. 5. Attack vs. non-attack transactions by event type

References:

[1] CNCF whitepaper: Serverless overview. URL: https://github.com/cncf/wg-serverless/blob/master/whitepapers/serverless-overview/cncf_serverless_whitepaper_v1.0.pdf, accessed on 13/09/2024

[2] Mora H, Mora-Gimeno FJ, Jimeno-Morenilla A., Macia-Lillo A., Elouali A. (2022) Serverless computing at the edge for AIoT applications. International Conference on Artificial Intelligence of Things (ICAIoT). DOI: 10.1109/ICAIoT57170.2022.10121879

[3] E. Jonas, A. Khandelwal and K. Krauth. *Cloud Programming Simplified: A Berkeley View on Serverless Computing.* UC Berkeley, 2019.

[4] State of serverless, DataDog, 2024. URL https://www.datadoghq.com/state-of-serverless/, accessed on 13/09/2024

[5] Candel, J.M.O., Elouali, A., Gimeno, F.J.M., Mora, H. (2023). Cloud vs Serverless Computing: A Security Point of View. Proceedings of the International Conference on Ubiquitous Computing & Ambient Intelligence (UCAmI 2022). UCAmI 2022. Lecture Notes in Networks and Systems, vol 594. Springer, Cham.

[6] Maciá-Lillo, A., Mora, H., Ramírez-Gordillo, T., Jimeno-Morenilla, A. (2024). Serverless Edge Providers for AI Applications. In: Visvizi, A., Troisi, O., Corvello, V. (eds) Research and Innovation Forum 2023. RIIFORUM 2023. Springer Proceedings in Complexity. Springer, Cham.

[7] Candel, J.M.O., Gimeno, F.J.M., Mora Mora, H. (2023). Serverless Security Analysis for IoT Applications. In: Bravo, J., Ochoa, S., Favela, J. (eds) Proceedings of the International Conference on Ubiquitous Computing & Ambient Intelligence (UCAmI 2022). UCAmI 2022. Lecture Notes in Networks and Systems, vol 594. Springer, Cham.

[8] Candel, J.M.O., Gimeno, F.J.M., Mora Mora, H. (2024) Generation of a dataset for DoW attack detection in serverless architectures, Data in Brief 52, 109921. DOI: 10.1016/j.dib.2023.109921
[9] Kelly D., Glavin F.G., Barrett E., (2023) DoWTS – Denial-of-wallet test simulator: synthetic data generation for preemptive defence, J. Intell. Inf. Syst., 60 (2023), pp. 325-348, DOI: 10.1007/s10844-022-00735-3
[10] H. C. Shin, K. Jameson, L. Matthew. (2018) Medical Image Sythesis for Data Augmentation and Anonymization using generative adversarial networks. Simulation and Synthesis in Medical Imaging, vol. 11037.
[11] S. Xun, H. Zhu, M. Chen. (2022) Generative adversarial networks in medical image segmentation: a review. Computers in biology and medicine. vol. 140.
[12] A. Biswas, A. Abdullah, A. Imran. (2023) Generative adversarial networks for data augmentation. Data Driven Approaches on Medical Imaging, pp. 159-177.
[13] N. Peppes, T. Alexakis, E. Adamopoulo. (2023) A comparasion study of GAN architectures for malicous cyber-attack data detection. School of electrical and computer engineering, National Technical University of Athens.